

ZENIVA: AYURVEDIC AI CLINICAL ASSISTANT

PROJECT DOCUMENTATION & TECHNICAL REPORT

1. Executive Summary

ZENIVA is an advanced multimodal, voice-enabled **Ayurvedic AI Clinical Assistant** designed to bridge classical Ayurvedic medical science with state-of-the-art Artificial Intelligence. The system utilizes **Google Gemini LLM**, **Retrieval-Augmented Generation (RAG)** with vector embeddings, real-time **Speech-to-Text (STT)**, continuous **Text-to-Speech (TTS)**, and an interactive **3D Holographic Command Dashboard** to provide personalized doshic assessment, patient EHR lookups, doctor discovery, and emergency triage.

2. System Architecture & Key Modules

- **Multilingual Speech-to-Text (STT):** Real-time Web Speech Recognition supporting Marathi (mr-IN), Hindi (hi-IN), and English with automatic silence detection and dispatch.
- **Text-to-Speech (TTS) & Talking Hologram Avatar:** Sentence-chunked female voice synthesis with a background keepalive watchdog preventing browser 15s freeze bugs. Synced with avatar lip-sync motion and dancing equalizer soundwaves.
- **Intelligent LLM with Zero-Downtime Fallback:** Dual-engine fallback cascading between gemini-3.5-flash-lite and gemini-3.6-flash with Vision capability for tongue and clinical image diagnosis.
- **Ayurvedic RAG Knowledge Base:** Powered by gemini-embedding-001 and Cosine Similarity across 11 core Ayurvedic knowledge modules (Tridosha, Agni, Ama, Prakriti, Vikriti, Dinacharya, etc.) plus custom PDF/CSV upload indexing.
- **2,000 Patient EHR Records (Electronic Health Records):** Indexed patient database (P0001–P2000) providing Prakriti, Vikriti, Tridosha balance, Agni fire metrics, Ama toxin levels, and customized dietary guidance.
- **200 Verified Ayurvedic Doctors & 1-Click Booking:** Search and filter 200 doctors across 10 cities and 5 specializations (Kayachikitsa, Panchakarma, Dietetics, Yoga, Digestive Health) with instant appointment confirmation.
- **Emergency Red-Flag Guardrail:** Real-time triage detecting critical acute symptoms (e.g., severe acute chest pain) and directing patients to immediate hospital care (Dial 108 / 112).

3. Technical Stack & Implementation

Component	Technologies & Libraries Used
Programming Languages	Python 3.11, JavaScript (ES6+), HTML5, CSS3
Backend Framework	Flask (Python Web Server), Python-Dotenv
AI & Language Model	Google Gemini API (gemini-3.5-flash-lite, gemini-3.6-flash, Vision)
RAG & Vector Search	gemini-embedding-001, Scikit-learn (Cosine Similarity), NumPy, PyPDF
Voice Pipeline	Web Speech API (webkitSpeechRecognition), Web SpeechSynthesis API
Frontend & UI/UX	Glassmorphism, CSS3 Keyframe Animations (Orbital Pod), FontAwesome 6
Database & Storage	JSON / CSV Databases (data/ folder), Python zipfile

4. Clinical Datasets Specification

Dataset Name	Records	Key Attributes & Purpose
Patients EHR Database	2,000	Patient_ID, Name, Age, Prakriti, Vikriti, Dosha, Agni, Ama, Diet, Sleep, Stress
Doctors Directory	200	Doctor_ID, Name, Qualification, Specialization, Experience, Languages, Timing, City
Ayurvedic Knowledge Base	11 Modules	Tridosha, Vata, Pitta, Kapha, Prakriti, Vikriti, Agni, Ama, Dinacharya, Jihva Pariksha
Clinical Conversations	2,000	Multi-intent classified interactions (diet, appointment, symptoms, doctor, emergency)

5. Presentation & Viva Pitch

'ZENIVA is a voice-enabled, multimodal Ayurvedic AI clinical assistant that combines Google Gemini LLM, RAG knowledge retrieval, 2,000 patient EHRs, and 200 Ayurvedic doctors into a futuristic holographic dashboard for intelligent, personalized healthcare guidance.'